Mashiro Okada

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RESEARCH INTEREST

I am a postdoctoral researcher with extensive experience in high-throughput phenotyping using UAVs, statistical modeling, and crop growth prediction. My goal is to transform agricultural practice through predictive, data-driven approaches. Over the past six years, I have focused on soybean, integrating UAV-remote sensing, simulations with 3D data, and genomic/statistical analyses to capture complex phenotype and forecast growth. My work bridges fundamental phenotyping and genetic analysis with practical decision support. I am particularly interested in combining these approaches with agricultural robotics and AI systems to achieve automated, efficient, and scalable crop production.

EDUCATION

The University of Tokyo, Tokyo, Japan

- Ph.D. in Biometry — Apr 2022 – Mar 2025
  - Dissertation: "A Study on the Prediction of Soybean Canopy Structure and the Modeling of Its Weed Suppression Effects for Improving Breeding and Cultivation Management Efficiency"
  - Supervisor: Prof. Hiroyoshi Iwata
  - Committee: Drs. Gota Morota, Naohiro Aoki, Wei Guo, and Yoshihiro Ohmori
  - Focus: High Throughput Phenotyping, Statistical Modeling, UAV-Remote Sensing, Simulation with 3D data.
- M.S. in Biometry — Apr 2020 – Mar 2022
  - Thesis: "Development of a Model for Estimating Soybean Aboveground Traits from Remotely Sensed Images Using Deep Learning and Its Application to Genetic Analysis"
  - Supervisor: Prof. Hiroyoshi Iwata
  - Committee: Drs. Yoshihiro Ohmori and Yuya Fukano
  - Focus: High Throughput Phenotyping, Image Analysis, UAV-Remote Sensing, Genetic analysis.

Shinshu University, Matsumoto, Nagano, Japan

- B.S. in Plant Molecular Biology — Apr 2016 – Mar 2020
  - Thesis: "The quantitative trait of *Brassica rapa* and environmental effect in fields"
  - Supervisor: Prof. Nobuaki Hayashida
  - Focus: Plant Molecular Breeding.

PROFESSIONAL EXPERIENCE

Laboratory of Field Phenomics,
Institute for Sustainable Agro-ecosystem Service,
Graduate School of Agricultural and Life Sciences,
The University of Tokyo, Tokyo, Japan

- Postdoctoral Researcher — Apr 2025 – **Present**
  - Project: Optimization of Weed Control in Soybean Field (Collaborative project with KUBOTA)
  - Project: Generation of Digital Twin of Forest for Evaluation of Carbon Credit
  - Supervisor: Prof. Wei Guo
  - Focus: High Throughput Phenotyping, Statistical Modeling, UAV-Remote Sensing, Simulation with 3D data.

Laboratory of Biomatry and Bioinformatics,
Department of Agricultural and Environmental Biology,
Graduate School of Agricultural and Life Sciences,
The University of Tokyo, Tokyo, Japan

- Research Assistant — Aug 2022 – Mar 2025
  - Project: Optimization of Weed Control in Soybean Field (Collaborative project with KUBOTA)
  - Supervisor: Prof. Hiroyoshi Iwata
  - Focus: High Throughput Phenotyping, Statistical Modeling, UAV-Remote Sensing, Simulation with 3D data.

Taisho University, Tokyo, Japan

- Research Assistant — May 2022 – Mar 2025
  - Project: Organized IoT-acquired weather data and developed analysis frameworks to help people with limited data experience utilize the data effectively for a flood management project.
  - Supervisor: Prof. Naoya Furuta

Kazusa DNA Research Institute, Kisarazu, Chiba, Japan

- Curator — Nov 2021 – Mar 2025
  - Project: Curated genomic data for Plant GARDEN, reading and analyzing over 500 published papers to contribute significantly to database construction and quality.
  - Supervisors: Drs. Sachiko Isobe and Hisako Ichihara

PUBLICATIONS

RESEARCH ARTICLE

2. **M. Okada**, C. Barras, Y. Toda, K. Hamazaki, Y. Ohmori, Y. Yamasaki, H. Takahashi, H. Takanashi, M. Tsuda, M. Y. Hirai, H. Tsujimoto, A. Kaga, M. Nakazono, T. Fujiwara, and H. Iwata, "High-throughput Phenotyping of Soybean Biomass: Conventional Trait Estimation and Novel Latent Feature Extraction Using UAV Remote Sensing and Deep Learning Models", *Plant Phenomics*, vol. 6, 0244, Sep 2024, DOI: 10.34133/plantphenomics.0244. [Press Release in Japanese]
1. S. Kenta, M. S. Abe, D. Kumakura, S. Nakaoka, F. Fujiwara, H. Miyamoto, T. Nakaguma, **M. Okada**, K. Sakurai, S. Shimizu, H. Iwata, H. Masuya, N. Nihei, and Y. Ichihashi, "Chemical-Mediated Microbial Interactions Can Reduce the Effectiveness of Time-Series-Based Inference of Ecological Interaction Networks", *International Journal of Environmental Research and Public Health*, no. 3: 1228, Jan 2022, DOI: 10.3390/ijerph19031228.

OTHER

1. H. Kakui, M. Minamikawa, M. Okada, S. Sakuma, **M. Okada**, K. Masuda, and T. Komatsuda, "The latest on study and research abroad during mid- and post-COVID-19 periods", *Breeding Science*, vol. 26, issue. 1, pp:85-90, Jun 2024, DOI: 10.1270/jsbbr.26.W06.

CONFERENCES

ORAL

14. P. Sanghera, **M. Okada**, J. Burridge, S. Kimura, H. Iwata, and W. Guo, "Plant Detection And Position Assignment In Mixed-Genotype Soybean Stands", The 16th International Conference of Asia-Pacific Federation for Information Technology in Agriculture (APFITA 2026), Taipei, Taiwan, Nov 2026
13. C. C. Chiang, Y. Imachi, **M. Okada**, S. Kamikawa, K. Tanaka, M. Miyauchi, K. Miyamoto, H. Iwata, J. Burridge, and W. Guo, "Potato Stem Counting from UAV and Ground-Level Images and Its Relationship with Tuber-Size Distribution", The 16th International Conference of Asia-Pacific Federation for Information Technology in Agriculture (APFITA 2026), Taipei, Taiwan, Nov 2026
12. **M. Okada**, and Wei Guo, "UAV Field Phenotyping Protocol: Toward Next-Generation Phenotyping from Field Mapping to Adaptive Observation", Japanese Society of Breeding 150th, Fukui Prefectural University, Fukui, Japan, Sep 2026
11. **M. Okada**, K. Miyamoto, M. Miyauchi, S. Kamikawa, D. Ishizuka, K. Yatsuda, J. Burridge, Wei Guo, and H. Iwata, "High Resolution Estimation of Sub-Canopy PAR for Low-Input Weed Management in Soybean", CIGR-EurAgEng World Congress 2026, Turin, Italy, Jun 2026
10. P. Sanghera, **M. Okada**, J. Burridge, S. Kimura, H. Iwata, and W. Guo, "Quantifying Dynamic Interplant Competition In Mixed Genotype Soybean Plots Using Traits Derived From UAV", CIGR-EurAgEng World Congress 2026, Turin, Italy, Jun 2026, *Note*: The abstract was accepted but it was not presented in the conference due to presenter's visa rejection.
9. S. Chen, H. Wang, **M. Okada**, X. Zhang, S. Kamikawa, N. Morinaga, and W. Guo, "A UAV-based Pipeline for Broccoli Head Size Estimation Across Varieties and Occlusion Conditions ", CIGR-EurAgEng World Congress 2026, Turin, Italy, Jun 2026
8. S. Chen, H. Wang, **M. Okada**, S. Kimura, R. Nizuguchi, N. Haramoto, Y. Ishizaki, H. Aoki, S. Kamikawa, W. Guo, "UAV-Based Estimation of Broccoli Head Size under Leaf Occlusion Using Time-Series Data", Japanese Society of Agricultural Informatics 2026, Kinki University, Osaka, Japan, May 2026
7. C. C. Chiang, Y. Imachi, **M. Okada**, S. Kamikawa, K. Tanaka, M. Miyauchi, K. Miyamoto, H. Iwata, J. Burridge, and W. Guo, "Development of a High-Throughput Phenotyping Pipeline Using Proximal Sensing for Time-Series Potato Stem Counting", Japanese Society of Agricultural Informatics 2026, Kinki University, Osaka, Japan, May 2026
6. H. Isagawa, K. Sakurai, S. Kinoshita, L. Jiaqi, T. Ishii, H. Takanashi, **M. Okada**, Y. Fuji, Y. Ohmori, H. Tanahashi, M. Tsuda, W. Guo, Y. Ichihashi, H. Iwata, and M. Hirai, "Assessment of Crop Interactions in Soybean-Sorghum Intercropping Using Time-Series UAV Data", Japanese Society of Breeding 148th, 410, Hokkaido University, Sapporo, Hokkaido, Japan, Sep 2025

5. **M. Okada**, K. Miyamoto, M. Miyauchi, S. Kamikawa, J. Burridge, W. Guo, and H. Iwata, “Spatio-Temporal Simulation of Canopy Light Environments in Soybean Fields Using UAV Remote Sensing and Ray Tracing”, PhenomUK 3rd Annual Conference, Coventry, UK, Sep 2025
4. **M. Okada**, R. Konishi, K. Miyamoto, M. Miyauchi, S. Kamikawa, J. Burridge, W. Guo, and H. Iwata, “Construction of soybean growth model to achieve efficient weed control based on UAV-remote sensing and no-linear mixed effect model”, Session 4 Lightning talk, International Plant Phenotyping Symposium 8, Lincoln, Nebraska, USA, Oct 2024 [YouTube]
3. Y. Toda, K. Hamazaki, **M. Okada**, K. Sakurai, Y. Fuji, Y. Ohmori, Y. Yamasaki, H. Takahashi, H. Takanashi, M. Tsuda, H. Kanegae, M. Hirai, H. Tsujimoto, A. Kaga, M. Nakazono, T. Fujiwara, Y. Ichihashi, and H. Iwata, “Comprehensive genome-wide association study of multi-omics data for soybean drought tolerance”, Japanese Society of Breeding 141st, 527, Online, Mar 2022
2. **M. Okada**, C. Barras, Y. Toda, Y. Ohmori, Y. Yamasaki, H. Takahashi, H. Takanashi, M. Tsuda, M. Hirai, H. Tsujimoto, A. Kaga, M. Nakazono, T. Fujiwara, and H. Iwata, “Genetic analysis of biomass-related features extracted by deep learning of UAV remote sensing images”, Japanese Society of Breeding 139th, 221, Online, Mar 2021
1. **M. Okada**, C. Barras, Y. Toda, Y. Ohmori, Y. Yamasaki, H. Takahashi, H. Takanashi, M. Tsuda, M. Hirai, H. Tsujimoto, A. Kaga, M. Nakazono, T. Fujiwara, and H. Iwata, “Deep learning for prediction of soybean biomass based on UAV remote sensing images”, Japanese Society of Breeding 138th, 514, Online, Oct 2020

POSTER

5. **M. Okada**, R. Konishi, K. Miyamoto, M. Miyauchi, S. Kamikawa, J. Burridge, W. Guo, and H. Iwata, “Construction of soybean growth model to achieve efficient weed control based on UAV-remote sensing and no-linear mixed effect model”, International Plant Phenotyping Symposium 8, Lincoln, Nebraska, USA, Oct 2024 (**Selected as top five poster in session 4; Next level AI, Models, and Algorithms for Phenomics**)
4. T. Chen, S. Kinoshita, **M. Okada**, Y. Imachi, K. Sakurai, T. Tsusaka, M. Sakurai, K. Shirasawa, S. Isobe, and H. Iwata, “Development of an objective evaluation method for the curling of perilla leaves using a convolutional neural network”, Japanese Society of Breeding 146th, P104, Hiroshima University, Hiroshima, Japan, Sep 2024
3. Y. Kobayashi, K. Seki, H. Takamori, **M. Okada**, K. Komatsu, K. Tanaka, Y. Uno, O. Watanabe, S. Nishiuchi, H. Matsumura, and N. Hayashida, “Evaluation of phenotypes using HTP technologies suggests effectivity to the investigation of genes”, Taiwan-Japan Plant Biology 2023, Taipei, Taiwan, Oct 2023
2. H. Takamori, K. Seki, Y. Kobayashi, **M. Okada**, K. Komatsu, K. Tanaka, Y. Uno, O. Watanabe, S. Nishiuchi, H. Matsumura, N. Hayashida, “QTL analysis of rosette size in tuberous lettuce using drones”, Japanese Society of Breeding 142nd, P022, Obihiro University of Agriculture and Veterinary Medicine, Obihiro, Hokkaido, Japan, Sep 2022
1. H. Takamori, O. Kawaide, T. Sakaguchi, M. Nakazawa, N. Ito, A. Furukubo, M. Amaike, T. Ito, F. Azuhata, **M. Okada**, S. Chino, H. Matsumura, S. Niikura, N. Hayashida, “Resistant loci to physiological disorder cupping in Chinese Cabbage(*Brassica rapa* var. *Pekinensis*)”, 10th Asian Crop Science Association Conference, P01159, Online, Sep 2021

WORKSHOP

1. H. Kakui, M. Minamikawa, M. Okada, S. Sakuma, **M. Okada**, K. Masuda, and T. Komatsuda, “Research abroad during mid- and post-COVID-19 periods”, Japanese Society of Breeding 144th, W06, Kobe University, Kobe, Hyogo, Japan, Sep 2023

INVITED TALK

2. T. Chen, Y. Fukumoto, **M. Okada**, S. Fukumoto, M. Okada, W. Guo, and H. Iwata, “Growth Model-Based Fusion of Satellite and Drone Remote Sensing Data”, Fifth International Workshop on Machine Learning for Cyber-agricultural Systems (MLCA2023), 12, Sarabetsu, Hokkaido, Japan, Jul 2023
1. M. Weiss, F. Baret, R. Lopez-Lozano, B. de Solan, H. Zhang, B. Mateo-Herrera, W. Li, **M. Okada** et al., “Satellite remote sensing applications for plant phenotyping”, 2023 Young Scientists Forum on Plant Phenomics, Nanjing, China, Jun 2023

ACADEMIC SERVICE

REVIEWER FOR PEER-REVIEWED JOURNALS

- *Breeding Science* 2026 – **Present**
  Number of reviewed manuscripts: 2026 (1)
- *The Plant Phenome Journal* 2026 – **Present**
  Number of reviewed manuscripts: 2026 (1)
- *Plant Phenomics* 2023 – **Present**
  Number of reviewed manuscripts: 2026 (4), 2025 (4), 2024 (1), 2023 (1)

SESSION CHAIR IN CONFERENCES

- *CIGR–EurAgEng World Congress 2026* Jun 2026
  Topic 7 - AI-Driven Crop Phenotyping & Digital Twins

AWARDS & SCHOLARSHIPS

- PhenomUK 3rd Annual Conference bursary recipient Sep 2025
  PhenomUK provide conference bursary up to £250 for early career scientists and students.
- Fully funded student, Tamagawa Academy-K12 (Machida, Tokyo, Japan) and The Essington School (Darwin, Northern Territory, Australia) Jul 2014
  For Tetsuro Obara tropical environmental program.

PROFESSIONAL MEMBERSHIPS

- Japanese Society of Breeding 2020 – **Present**

OTHER ACADEMIC ACTIVITY

- Technical Training Session Aug 2026
  - AI Tools and Techniques for Developing a Database of Medical Plants with Antimicrobial Activity
    Organizer: Himalayan Environment and Public Health Network (HEPHN)
    Role: Co-instructor
    Lead instructor: Prof. Wei Guo
- Research Exchange Feb 2023 – May 2023
  - CAPTE Research Group, EMMAH Unit, INRAE, Avignon, France
    Host researchers: Benoit de Solan and Dr. Marie Weiss.
    Financially supported by JSPS KAKENHI (Grant No. 22K21352).

INVITED PRESENTATION

1. **M. Okada**, Super Science Highschool Special Lecture, Tamagawa Academy K-12, Machida, Tokyo, Japan, Jun 2023

LANGUAGES

- Japanese: Native
- English: Fluent (speaking, reading, writing)

SKILLS

OPERATING SYSTEM

- Linux (Ubuntu), Mac OSX, Windows

PROGRAMMING

- R, Python, TEX, LATEX, bash

COMPUTER VISION

- OpenCV, Open3D, Metashape, NeRF Studio

GENETIC ANALYSIS

- Genomic Prediction, GWAS, QTL-analysis

DEEP LEARNING

- Pytorch, Keras, YOLO, SAMGeo

OTHER

- Driving license (2016), UAV flight (Imaging and LiDAR scanning)

REFERENCES

- **Professor Wei Guo**
  Principal Investigator of Laboratory of Field Phenomics
  The University of Tokyo
  1-1-1 Midori-cho, Nishitokyo, Tokyo, 188-0002, Japan
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▪ **Professor Hiroyoshi Iwata**
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[*CV compiled on 2026-10-28*]